

ctrl + arm

Say it. The arm scans it in 3D.

THE PROBLEM

3D scanning still means a person walking in circles.

01

By hand

One angle at a time, camera in hand.

02

Every shot is a guess

Blur, holes, no real scale.

03

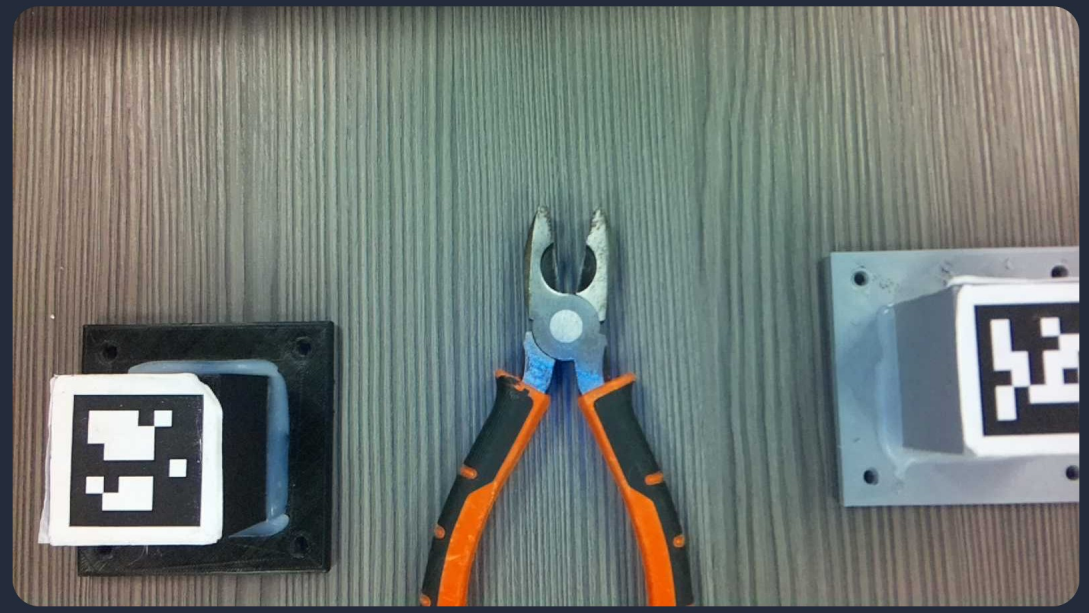
It does not scale

One object, one expert.

THE IDEA

Put the camera on a robot arm. Let Grok drive it by voice.

The arm knows where every photo was taken: exactly what a 3D model needs.



The pliers, as the arm camera sees them

HOW IT WORKS

From a sentence to a mesh

01
Listen
Grok Voice

02
See
Grok picks the object

03
Move
12 views around it

04
Capture
Pi camera, 10 fps

05
Build
Mesh in millimetres

Grok decides what you mean. Code decides where the arm goes.

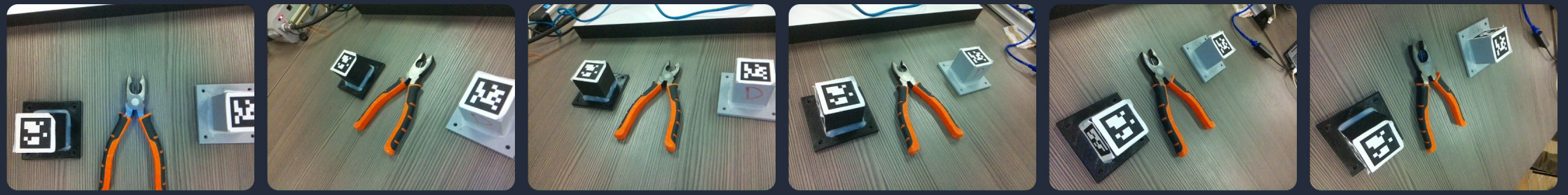
VOICE

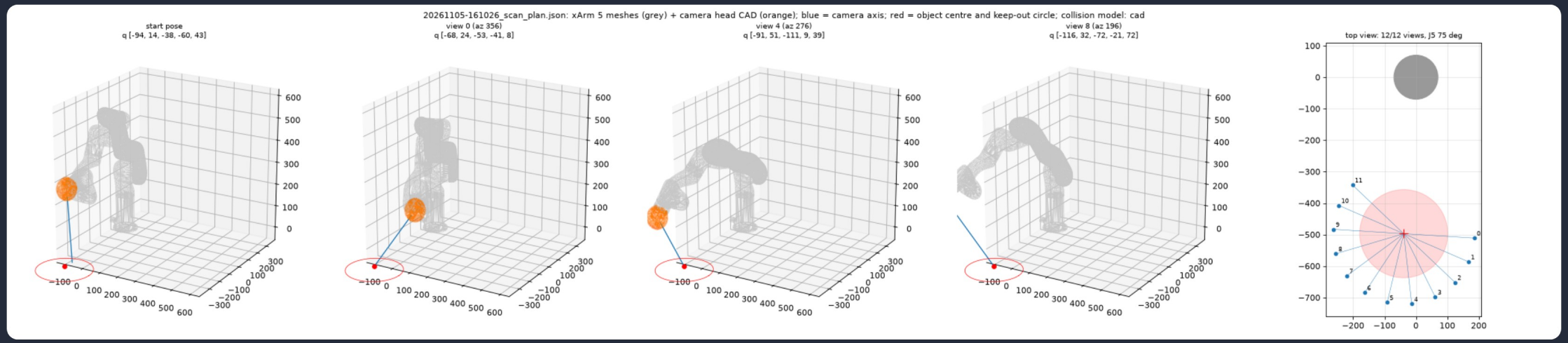
Every step is a sentence

You say	In English	What happens
“Scansiona il tavolo”	Scan the table	Turns 360°: 12 photos, a live map of every object
“Trovami le pinze”	Find me the pliers	Finds them on the map and asks “Confermi?”
“Sì”	Yes	Goes over them and takes a photo from above
“Scansionalo”	Scan it	Plans the next view; every move is confirmed
“L’ho capovolto”	I flipped it	Starts a new side of the object
“Ricostruiscilo”	Rebuild it	Runs the reconstruction and returns the mesh
“Stop” · “Fermo”	Stop	The arm holds at once. It never waits for the model.

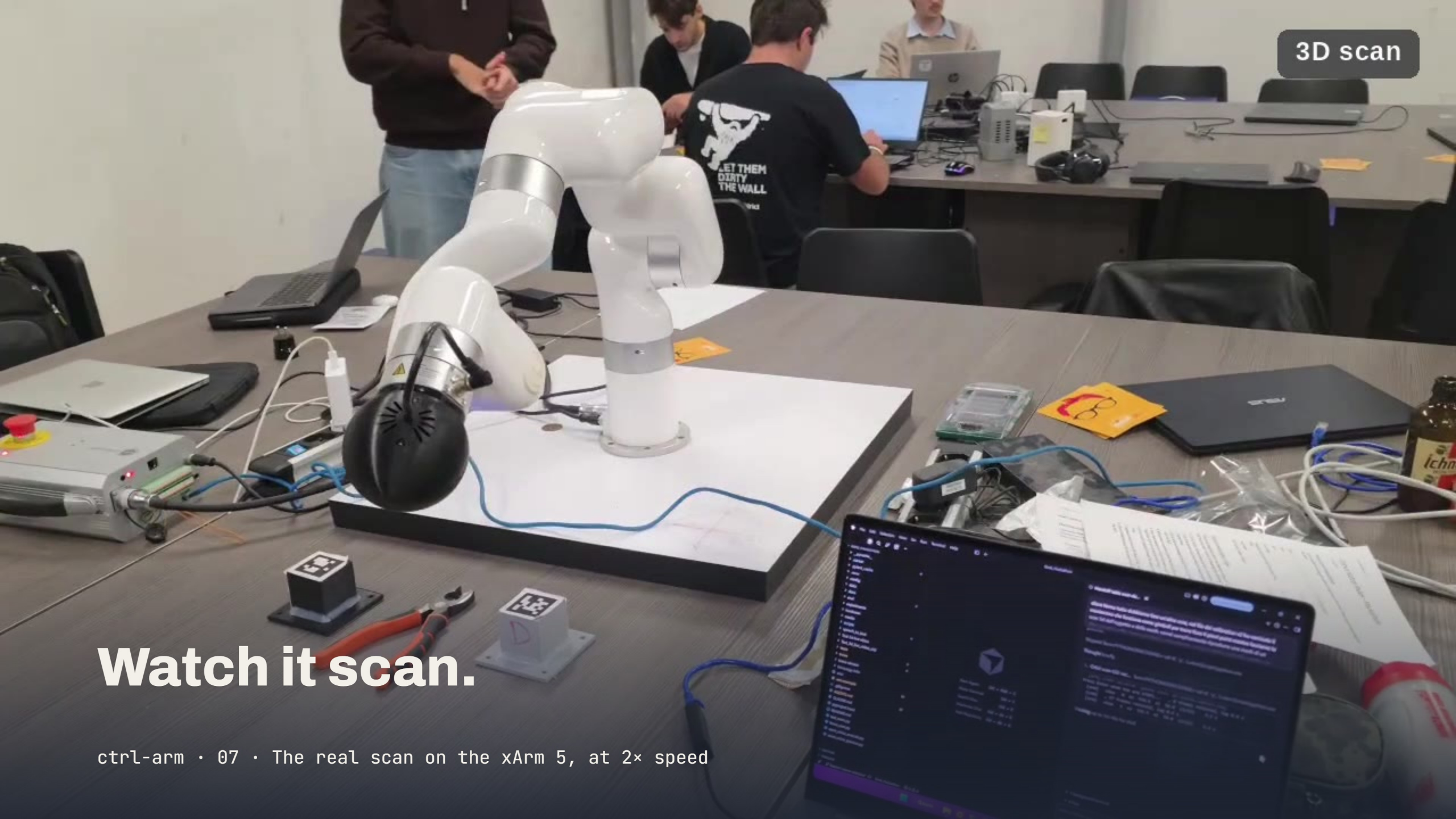
ONE ORBIT

81 photos around the pliers





ctrl-arm · 06 · Real frames from the Pi camera · orbit plan checked on the xArm 5 CAD model



Watch it scan.

ctrl-arm · 07 · The real scan on the xArm 5, at 2× speed

BUILT IN 36 HOURS

What runs today, on the real xArm 5

84 objects

In one 360° table scan

12 photos, about a minute

81 photos

In one orbit of the pliers

Pi camera at 10 fps

180.7 mm

Pliers length, on the mesh

L 180.7 · W 99.0 · H 18.1 mm

1.4 mm

Median calibration error

Camera to robot, recorded data

6.0 mm

Worst-case error (p95)

Down from 13.5 mm, same images

281 tests

Passing

Simulated arm and camera

IT DID NOT WORK AT FIRST

Nothing worked. Then, at night, it did.

MORNING	**Arm frozen.** Reset the controller.
NOON	**Camera gone.** Reset the Raspberry Pi.
NIGHT	**No way to reach the arm.** Built our own LAN.
NIGHT	**First 3D scan: the cup.**
5 NOV 12:59	**The camera hit the arm.** Wrist limited, pose taught by hand.
5 NOV 13:31	**84 objects in one scan.**



Early mesh: the red mouse
Mesh 95.6 × 57.5 × 35.2 mm
Ruler 98 × 58 × 35 mm

Flip it.

NEXT · CONCEPT

Hang the arm from the ceiling.

- Every side of the desk, from above.
- Nothing on your table.
- It scans while you work.

SCALE THE SENSORS

Bolt on any sensor.

BUILT

RGB

Pi camera, 10 fps

NEXT

Depth

Shape without texture

NEXT

LiDAR

Metric, at room scale

NEXT

+1 arm

Turns the object for you

The arm is the constant. Every reading comes with a known pose, so everything fuses into one model.

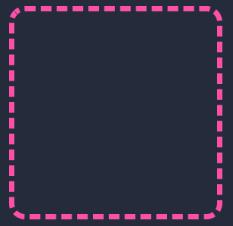
SCALE THE PLACE

Same voice. Any size.



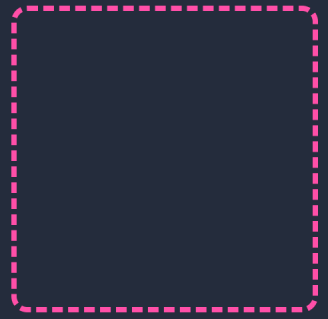
Desk

TODAY · one arm



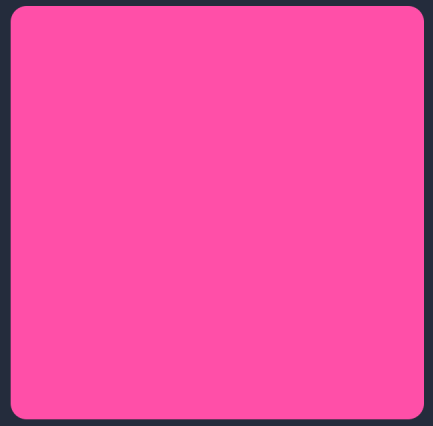
Room

NEXT · a ceiling arm



Building

NEXT · arms on wheels, robot dogs



Site

VISION · a fleet, one 3D map

FOR CREATIVES

Throw it on the table. Get the CAD on your PC.



→

“Scan it.”

→

STL GLB PLY

in real millimetres

Drop it

Any prop, on the table

Say it

In Italian or English

Open it

Blender, Unity, a 3D printer

Fast by design: 10 fps capture, GPU reconstruction, no scanning skills needed.

WHERE IT MATTERS MOST

After an earthquake: map the rubble, find the way through, move what blocks it.

Map

Cameras and LiDAR on arms, robot dogs or drones build a 3D model of the debris.

Find a path

Grok reads the model: where can a rescuer or a robot pass?

Clear it

The arm moves the obstacle in the way. A person confirms every move.

ctrl + arm

Say it. Scan it. Anywhere.

github.com/Aman-AI-Guy/CTRL-ARM



Final mesh · L 180.7 × W 99.0 × H 18.1 mm

Giacomo Alessandro Ferrari
ferrari.giacomoalessandro@gmail.com

Bhoumik Shankar Mishra
bhoumikmishra@gmail.com

Simone Stocco
simone.stocco01@universitadipavia.it

Nawaz Ahmad Karimi
nawazahmad6200@gmail.com

Aman Sahu
amansahuofficial11@gmail.com

Saurav Shriwastav
s.saurav@pcdue.com